

SERA FREQUENCY UTILIZATION PLAN for 420-450 MHz	
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Some Definitions:

Narrow-band FM transmission is 2.5 kHz maximum deviation consuming 11 to 12.5 kHz of bandwidth

Wide-band FM transmission is 5 kHz maximum deviation consuming 13 to 16 kHz of bandwidth

Coordination parameter tags:

NBD - Narrow-band digital – DStar, NXDN, P25, DMR, 2.5kHz analog, 2.5kHz analog in mixed-mode

WBD - Wide-band digital – Fusion (must be coordinated on wide-band pair)

WBA - Wide-band analog – on designated wide-band pairs, 5.0kHz analog, 5.0kHz analog in mixed-mode

Narrow Band FM Digital – NXDN / Dstar / DMR / P25 / NBFM Analog 2.5 kHz

Packet Repeater or Digital Packet / Duplex Backbone

18 Duplex Channels, 12.5 kHz channel spacing, 5 MHz split, High input – Low output, (NBD)

445.5125 – 440.5125	445.5250 – 440.5250	445.5375 – 440.5375
445.5500 – 440.5500	445.5625 – 440.5625	445.5750 – 440.5750
445.5875 – 440.5875	445.6000 – 440.6000	445.6125 – 440.6125
445.6250 – 440.6250	445.6375 – 440.6375	445.6500 – 440.6500
445.6625 – 440.6625	445.6750 – 440.6750	445.6875 – 440.6875
445.7000 – 440.7000	445.7125 – 440.7125	445.7250 – 440.7250

Standard FM Repeaters – Analog / NXDN / Fusion / P25 / DMR / Mixed Mode wide band FM

128 Repeater Pairs, channel spaced at 25 kHz intervals and interlaced frequency spacing at 12.5 kHz (active frequencies are 12.5 kHz apart.) Maximum Deviation limited to wide-band FM (WBA) (WBD) (NBD). High input – Low output

446.8000 – 441.8000	447.6000 – 442.6000	448.4000 – 443.4000	449.2000 – 444.2000
446.8250 – 441.8250	447.6250 – 442.6250	448.4250 – 443.4250	449.2250 – 444.2250
446.8500 – 441.8500	447.6500 – 442.6500	448.4500 – 443.4500	449.2500 – 444.2500
446.8750 – 441.8750	447.6750 – 442.6750	448.4750 – 443.4750	449.2750 – 444.2750
446.9000 – 441.9000	447.7000 – 442.7000	448.5000 – 443.5000	449.3000 – 444.3000
446.9250 – 441.9250	447.7250 – 442.7250	448.5250 – 443.5250	449.3250 – 444.3250
446.9500 – 441.9500	447.7500 – 442.7500	448.5500 – 443.5500	449.3500 – 444.3500
446.9750 – 441.9750	447.7750 – 442.7750	448.5750 – 443.5750	449.3750 – 444.3750
447.0000 – 442.0000	447.8000 – 442.8000	448.6000 – 443.6000	449.4000 – 444.4000
447.0250 – 442.0250	447.8250 – 442.8250	448.6250 – 443.6250	449.4250 – 444.4250
447.0500 – 442.0500	447.8500 – 442.8500	448.6500 – 443.6500	449.4500 – 444.4500
447.0750 – 442.0750	447.8750 – 442.8750	448.6750 – 443.6750	449.4750 – 444.4750
447.1000 – 442.1000	447.9000 – 442.9000	448.7000 – 443.7000	449.5000 – 444.5000
447.1250 – 442.1250	447.9250 – 442.9250	448.7250 – 443.7250	449.5250 – 444.5250
447.1500 – 442.1500	447.9500 – 442.9500	448.7500 – 443.7500	449.5500 – 444.5500
447.1750 – 442.1750	447.9750 – 442.9750	448.7750 – 443.7750	449.5750 – 444.5750
447.2000 – 442.2000	448.0000 – 443.0000	448.8000 – 443.8000	449.6000 – 444.6000
447.2250 – 442.2250	448.0250 – 443.0250	448.8250 – 443.8250	449.6250 – 444.6250
447.2500 – 442.2500	448.0500 – 443.0500	448.8500 – 443.8500	449.6500 – 444.6500
447.2750 – 442.2750	448.0750 – 443.0750	448.8750 – 443.8750	449.6750 – 444.6750
447.3000 – 442.3000	448.1000 – 443.1000	448.9000 – 443.9000	449.7000 – 444.7000
447.3250 – 442.3250	448.1250 – 443.1250	448.9250 – 443.9250	449.7250 – 444.7250
447.3500 – 442.3500	448.1500 – 443.1500	448.9500 – 443.9500	449.7500 – 444.7500
447.3750 – 442.3750	448.1750 – 443.1750	448.9750 – 443.9750	449.7750 – 444.7750
447.4000 – 442.4000	448.2000 – 443.2000	449.0000 – 444.0000	449.8000 – 444.8000
447.4250 – 442.4250	448.2250 – 443.2250	449.0250 – 444.0250	449.8250 – 444.8250
447.4500 – 442.4500	448.2500 – 443.2500	449.0500 – 444.0500	449.8500 – 444.8500
447.4750 – 442.4750	448.2750 – 443.2750	449.0750 – 444.0750	449.8750 – 444.8750
447.5000 – 442.5000	448.3000 – 443.3000	449.1000 – 444.1000	449.9000 – 444.9000
447.5250 – 442.5250	448.3250 – 443.3250	449.1250 – 444.1250	449.9250 – 444.9250
447.5500 – 442.5500	448.3500 – 443.3500	449.1500 – 444.1500	449.9500 – 444.9500
447.5750 – 442.5750	448.3750 – 443.3750	449.1750 – 444.1750	449.9750 – 444.9750

APCO P25, DMR, Dstar, NXDN, NBFM Analog Repeaters (Narrow band protocols)

128 Repeater Pairs, channel spaced at 25 kHz intervals and interlaced frequency spacing at 12.5 kHz (active frequencies are 12.5 kHz apart.) NBFM 2.5 kHz Analog. Maximum Deviation limited to established *digital* protocol standards (NBD). High input, Low output.

446.8125 - 441.8125	447.6125 - 442.6125	448.4125 - 443.4125	449.2125 - 444.2125
446.8375 - 441.8375	447.6375 - 442.6375	448.4375 - 443.4375	449.2375 - 444.2375
446.8625 - 441.8625	447.6625 - 442.6625	448.4625 - 443.4625	449.2625 - 444.2625
446.8875 - 441.8875	447.6875 - 442.6875	448.4875 - 443.4875	449.2875 - 444.2875
446.9125 - 441.9125	447.7125 - 442.7125	448.5125 - 443.5125	449.3125 - 444.3125
446.9375 - 441.9375	447.7375 - 442.7375	448.5375 - 443.5375	449.3375 - 444.3375
446.9625 - 441.9625	447.7625 - 442.7625	448.5625 - 443.5625	449.3625 - 444.3625
446.9875 - 441.9875	447.7875 - 442.7875	448.5875 - 443.5875	449.3875 - 444.3875
447.0125 - 442.0125	447.8125 - 442.8125	448.6125 - 443.6125	449.4125 - 444.4125
447.0375 - 442.0375	447.8375 - 442.8375	448.6375 - 443.6375	449.4375 - 444.4375
447.0625 - 442.0625	447.8625 - 442.8625	448.6625 - 443.6625	449.4625 - 444.4625
447.0875 - 442.0875	447.8875 - 442.8875	448.6875 - 443.6875	449.4875 - 444.4875
447.1125 - 442.1125	447.9125 - 442.9125	448.7125 - 443.7125	449.5125 - 444.5125
447.1375 - 442.1375	447.9375 - 442.9375	448.7375 - 443.7375	449.5375 - 444.5375
447.1625 - 442.1625	447.9625 - 442.9625	448.7625 - 443.7625	449.5625 - 444.5625
447.1875 - 442.1875	447.9875 - 442.9875	448.7875 - 443.7875	449.5875 - 444.5875
447.2125 - 442.2125	448.0125 - 443.0125	448.8125 - 443.8125	449.6125 - 444.6125
447.2375 - 442.2375	448.0375 - 443.0375	448.8375 - 443.8375	449.6375 - 444.6375
447.2625 - 442.2625	448.0625 - 443.0625	448.8625 - 443.8625	449.6625 - 444.6625
447.2875 - 442.2875	448.0875 - 443.0875	448.8875 - 443.8875	449.6875 - 444.6875
447.3125 - 442.3125	448.1125 - 443.1125	448.9125 - 443.9125	449.7125 - 444.7125
447.3375 - 442.3375	448.1375 - 443.1375	448.9375 - 443.9375	449.7375 - 444.7375
447.3625 - 442.3625	448.1625 - 443.1625	448.9625 - 443.9625	449.7625 - 444.7625
447.3875 - 442.3875	448.1875 - 443.1875	448.9875 - 443.9875	449.7875 - 444.7875
447.4125 - 442.4125	448.2125 - 443.2125	449.0125 - 444.0125	449.8125 - 444.8125
447.4375 - 442.4375	448.2375 - 443.2375	449.0375 - 444.0375	449.8375 - 444.8375
447.4625 - 442.4625	448.2625 - 443.2625	449.0625 - 444.0625	449.8625 - 444.8625
447.4875 - 442.4875	448.2875 - 443.2875	449.0875 - 444.0875	449.8875 - 444.8875
447.5125 - 442.5125	448.3125 - 443.3125	449.1125 - 444.1125	449.9125 - 444.9125
447.5375 - 442.5375	448.3375 - 443.3375	449.1375 - 444.1375	449.9375 - 444.9375
447.5625 - 442.5625	448.3625 - 443.3625	449.1625 - 444.1625	449.9625 - 444.9625
447.5875 - 442.5875	448.3875 - 443.3875	449.1875 - 444.1875	449.9875 - 444.9875

FM Itinerant Voice Repeaters

2 Repeater Pairs, 25 kHz spacing between channels, 5 MHz split, High Input – Low Output

445.750 – 440.750

445.800 – 440.800

Reserved for itinerant and short term use during a special event or emergency with no formal coordination required. Trustees/Owners are encouraged to notify their State coordinator to inform them of system parameters and expected duration of use.

Non-Coordinated Repeater Channels (Backyard Repeater)

In addition, the “itinerant and short term” channels presently available in the 70 cm band, are intended to be portable or movable to different locations as needed. SERA will now allocate two additional channels 445.850 / 440.850 and 445.875 / 440.875 to be used for short-range repeater communications (backyard systems). These two pairs will be non-coordinated, stationary, and would have no protection against interference. These non-coordinated repeaters will consist of an antenna height of no more than 50' feet above ground and 50 feet above average terrain computed with the FCC Calculator. Power is limited to 50 watts ERP

Digital Hotspots for DMR, DStar, NXDN, P25, YSF, etc.

440.9250	440.9500	440.9750	441.0250	441.0500
441.0750	441.1000	441.1250	441.1500	441.1750

“Hotspots” are low-power digital interface devices intended to bridge between the Internet and hand-held digital-mode radios. These recommended frequencies are shared with Narrow Band FM Digital Simplex users, and care should be taken to not choose a frequency already used locally. No amplifier or external antenna should be used with a hotspot, and the transmit power of the corresponding radio should be set to minimum.

Narrow Band 12.5 kHz FM Digital Simplex, DStar / DMR / P25 (APC025) / NXDN / Packet / NBFM 2.5 kHz Analog

440.9125	440.9250	440.9375	440.9500	440.9625	440.9750
440.9875	441.0000 *	441.0125	441.0250	441.0375	441.0500
441.0625	441.0750	441.0875	441.1000	441.1125	441.1250
441.1375	441.1500	441.1625	441.1750		

* 441.000 is the National Digital Simplex Calling Frequency

Voice FM Simplex Frequencies – 22 Simplex FM Voice Channels

445.9125	445.9250	445.9375	445.9500	445.9625	445.9750
455.9875	446.0000 *	446.0125	446.0250	446.0375	446.0500
446.0625	446.0750	446.0875	446.1000	446.1125	446.1250
446.1375	446.1500	446.1625	446.1750		

* Note that 446.0000 is the National FM Voice Simplex Channel

FM Cross Band Repeat Simplex (Narrow band FM 12.5 kHz)

Channels for **Fixed** Stations ONLY, 12 Simplex Channels, for Fixed Cross Band Repeat

440.7375	440.7625	440.7750	440.7875	440.8125	440.8250
440.8375	440.8500	440.8625	440.8750	440.8875	440.9000

Channels for **Mobile** Stations ONLY, 12 Simplex Channels, for Mobile Cross Band Repeat

445.7375	445.7625	445.7750	445.7875	445.8125	445.8250
445.8375	445.8500	445.8625	445.8750	445.8875	445.9000

Link & Control Channels (Narrow Band FM 12.5 kHz)

May be paired on "440", or individually paired with 222 MHz band fixed link frequencies. These may be individually coordinated. Some may be coordinated as simplex remote base or remote control systems.

446.1875 - 441.1875	446.2000 - 441.2000	446.2125 - 441.2125
446.2250 - 441.2250	446.2375 - 441.2375	446.2500 - 441.2500
446.2625 - 441.2625	446.2750 - 441.2750	446.2875 - 441.2875
446.3000 - 441.3000	446.3125 - 441.3125	446.3250 - 441.3250
446.3375 - 441.3375	446.3500 - 441.3500	446.3625 - 441.3625
446.3750 - 441.3750	446.3875 - 441.3875	446.4000 - 441.4000
446.4125 - 441.4125	446.4250 - 441.4250	446.4375 - 441.4375
446.4500 - 441.4500	446.4625 - 441.4625	446.4750 - 441.4750
446.4875 - 441.4875		

Link Channels (Narrow Band FM 12.5 kHz)

These link channels share designated ATV Bandwidths using horizontal antenna polarity. Frequencies will be coordinated with a maximum 25 watts ERP and directional antenna systems.

420.0125	433.0125	438.7750	438.9750	439.1750 *	439.3875 *
420.0250	433.0250	438.7875	438.9875	439.1875 *	439.4000 *
420.0375	433.0375	438.8000	439.0000 *	439.2000 *	439.4125 *
420.0500	433.0500	438.8125	439.0125 *	439.2125 *	439.4250 *
420.0625	433.0625	438.8250	439.0250 *	439.2250 *	439.4375 *
420.0750	433.0750	438.8375	439.0375 *	439.2375 *	439.4500 *
420.0875	433.0875	438.8500	439.0500 *	439.2500 *	439.4625 *
420.1000	433.1000	438.8625	439.0625 *	439.2625 *	439.4750 *
420.1125	433.1125	438.8750	439.0750 *	439.2750 *	439.4875 *
420.1250	433.1250	438.8875	439.0875 *	439.2875 *	439.5000 *
420.1375	433.1375	438.9000	439.1000 *	439.3000 *	439.5125 *
420.1500	433.1500	438.9125	439.1125 *	439.3125 *	439.5250 *
420.1625	433.1625	438.9250	439.1250 *	439.3250 *	439.5375 *
420.1750	433.1750	438.9375	439.1375 *	439.3375 *	439.5500 *
420.1875	433.1875	438.9500	439.1500 *	439.3500 *	439.5625 *
420.2000	433.2000	438.9625	439.1625 *	439.3625 *	439.5750 *
				439.3750 *	439.5875 *

* Link frequencies above 439 MHz may use vertical polarity at the coordinator's discretion.

Duplex "Hub" Linking Repeaters

11 Repeater Pairs, 25 KHz spacing, 5 MHz split, High input, Low output, maximum 100w ERP.

446.525 - 441.525	446.550 - 441.550	446.575 - 441.575
446.600 - 441.600	446.625 - 441.625	446.650 - 441.650
446.675 - 441.675	446.700 - 441.700	446.725 - 441.725
446.750 - 441.750	446.775 - 441.775	

These repeater pairs are reserved for use as a wide area linking "hub" repeater that tie a group of standard FM voice repeaters together. The SERA strongly recommends this method of linking rather than using in band or remote base type linking.

Wide Band Digital / Packet – Duplex Backbones / Links

19 Duplex, Mixed 50/100 KHz, 5 MHz split, High input – Low output,
horizontal or vertical polarization optional

445.0250 – 440.0250	445.0500 – 440.0500	445.0750 – 440.0750
445.1000 – 440.1000	445.1250 – 440.1250	445.1500 – 440.1500
445.1750 – 440.1750	445.2000 – 440.2000	445.2250 – 440.2250
445.2500 – 440.2500	445.2750 – 440.2750	445.3000 – 440.3000
445.3250 – 440.3250	445.3500 – 440.3500	445.3750 – 440.3750
445.4000 – 440.4000	445.4250 – 440.4250	445.4500 – 440.4500
445.4750 – 440.4750		

Frequency Spectrum Plan for 420–450 MHz

Voice FM Simplex Frequencies

22 Simplex FM Voice Channels

445.9125 445.9250 445.9375 445.9500 445.9625 445.9750
445.9875 446.0000 * 446.0125 446.0250 446.0375 446.0500
446.0625 446.0750 446.0875 446.1000 446.1125 446.1250
446.1375 446.1500 446.1625 446.1750

446.0000 is the National FM Voice Simplex Calling Frequency

420.0000 – 420.2000 Repeater Links

420.0000 – 426.0000 ATV Repeater Output 421.250 Video – 425.750 Audio

426.0000 – 432.0000 ATV Simplex 427.250 Video – 431.750 Audio

432.0000 – 433.0000 SSB & Weak Signal 432.100 National SSB Calling Frequency

433.0125 – 433.2000 Repeater Links

433.0000 – 439.0000 ATV Simplex Only 434.000 Video – 438.500 Audio

438.7750 – 439.5875 Repeater Links

439.6000 – 440.4750 FM Digital/Packet Operation

440.5125 – 440.7250 Narrow Band Digital Repeater Outputs/Duplex Backbones/Link

440.7375 – 440.9000 Simplex Cross Band Voice – Base

440.9125 – 441.1750 Simplex Digital

441.1875 – 441.4875 Links, Control Frequencies

441.5000 Simplex Digital 441.000 National Digital Calling Frequency

441.5250 – 441.7750 Duplex Link Repeaters

441.8000 – 444.9750 FM, APCO P25, DMR, Fusion and D-Star Repeater Outputs

445.0250 – 445.4750 Digital

445.5125 – 445.7250 Digital Repeater Inputs

445.7375 – 445.9000 Simplex Cross Band Voice – Mobile

445.9125 – 446.1750 FM Simplex 446.000 National FM Calling Frequency

446.1875 – 446.4875 Links, Control Frequencies

446.5000 Simplex Digital

446.5250 – 446.7750 Duplex Link Repeaters

446.8000 – 449.9750 FM, APCO P25, DMR, Fusion and D-Star Repeater Inputs,
wideband/mixed mode digital

NOTE: *Contact your respective SERA Director, Vice Director, or Frequency Coordinator, for information about the need for coordination of your system, before installing and operating the system. Most simplex voice and digital-packet operations are itinerant in nature, and as such will not be coordinated by SERA. Digital-Packet operators should contact their local digital-packet organization for additional information. Simplex cross band voice, as is possible with most over the counter dual band radios, is also considered itinerant in nature and will not be coordinated; however, users are asked to use the assigned frequencies shown in this document.*

Version Updates:
Draft Version 02-24

